Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.13	20.20	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.02	+0.3%
power (W), mean	626	641	+2.5%
power (W), peak	646	682	+5.6%
energy (Wh)	210	216	+2.8%
energy, parked workers still on at idle power (Wh)	210	216	+2.8%
CPU used (cores), mean	0.803	1.21	+51.1%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.033	0.051	+51.1%
energy per core-hour (Wh)	779	528	-32.2%
node-hours per core-hour	7.47	4.94	-33.8%
pending pods, pod-minutes	0.267	0.267	+0.0%
pending pods, peak	1.00	1.00	+0.0%
HPA replicas, mean	8.71	9.42	+8.2%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.500	0.783	+56.7%
requests timed	1,125	1,100	-2.2%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	68.91	95.24	+38.2%
response time (ms), median	46.30	81.50	+76.0%
response time (ms), 95th percentile	166	153	-8.2%
response time (ms), 99th percentile	249	185	-25.6%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	8.2639	5.7187	-30.8%	[-4.5097, -0.5828]	GAIN
kwh_per_core_hour	0.8579	0.6054	-29.4%	[-0.4386, -0.0587]	GAIN
utilisation	0.0332	0.0496	+49.5%	[+0.0039, +0.0297]	GAIN
pending_pod_minutes_per_hour	0.7805	0.7619	-2.4%	[-2.3415, +2.2857]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	1.4634	2.2619	+54.6%	[-2.1951, +4.5302]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.358 0.392 0.421 0.446 0.468 0.488 0.505 0.519 0.533 0.546 0.555 0.561 0.567 0.571 0.576 0.581 0.584 0.588 0.589 0.59
- nodes decided per minute: 6 6 5 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	615	621	5.00	5.00	0.000	0.000	0.491	0.624

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	616	624	5.00	7.75	0.000	0.000	0.536	0.750
2	6.00	6.00	615	623	5.00	8.00	0.000	0.000	0.507	0.723
3	6.00	6.00	620	633	5.00	8.00	0.250	0.250	0.642	1.02
4	6.00	6.00	628	648	7.25	10.00	0.000	0.000	0.864	1.40
5	6.00	6.00	631	647	8.00	10.00	0.000	0.000	0.942	1.38
6	6.00	6.00	634	648	9.50	10.00	0.000	0.000	1.04	1.39
7	6.00	6.00	644	674	10.00	10.00	0.000	0.000	1.28	2.08
8	6.00	6.00	644	674	10.00	10.00	0.000	0.000	1.27	2.06
9	6.00	6.00	645	672	10.00	10.00	0.000	0.000	1.30	1.99
10	6.00	6.00	638	660	10.00	10.00	0.000	0.000	1.13	1.69
11	6.00	6.00	617	626	10.00	10.00	0.000	0.000	0.566	0.793
12	6.00	6.00	616	626	10.00	10.00	0.000	0.000	0.561	0.811
13	6.00	6.00	621	632	10.00	10.00	0.000	0.000	0.683	0.966
14	6.00	6.00	630	655	10.00	10.00	0.000	0.000	0.916	1.59
15	6.00	6.00	629	645	10.00	10.00	0.000	0.000	0.907	1.39
16	6.00	6.00	630	650	10.00	10.00	0.000	0.000	0.948	1.45
17	6.00	6.00	620	633	10.00	10.00	0.000	0.000	0.634	0.997
18	6.00	6.00	613	622	10.00	10.00	0.000	0.000	0.473	0.703
19	6.00	6.00	616	622	10.00	10.00	0.000	0.000	0.542	0.697

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.